Fiscal Year:	FY 2017	**Task Last Updated:**	FY 06/29/2018
PI Name:	Burke, Shawn Ph.D.		
Project Title:	Leadership-Followership: Moving Beyond Traditional Leadership to Build Highly Functioning Autonomous Teams		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32826-3281	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 HERO NNJ13ZSA002N-Crew Health OMNIBUS
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No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	5	**No. of Master' Degrees:**	1
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	7
No. of Bachelor's Candidates:	10	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date is 9/29/2017 per PI and NSSC information (Ed., 8/2/17) NOTE: End date changed to 7/31/2017 (original end date was 9/30/2015) per NSSC information (Ed., 3/20/17) NOTE: Element change to Human Factors & Behavioral Performance; previously Behavioral Health & Performance (Ed., 1/17/17) NOTE: End date changed to 3/31/2017 (original end date was 9/30/2015) per PI (Ed., 8/25/16) NOTE: End date changed to 1/29/2016 (original end date was 9/30/2015) per NSSC (Ed., 3/2/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Salas, Edward Ph.D. (University of Central Florida) Shuffler, Marissa Ph.D. (Clemson University)		
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Performance Goal Text:			

Task Description:	The long-duration of space exploration missions (and relative isolation) makes it imperative that teams have the leadership capacity to address the temporal dynamics of diversity and the rifts that can occur. Leadership has been shown to be a critical component of team effectiveness and adaptation. However, leadership may be even more important within autonomous teams as it serves to create the shared behavior, affect, and cognition that allows teams to adapt and be effective within complex environments. We argue that a particular form of leadership (i.e., shared or distributed leadership) is needed to produce highly functioning teams that are capable operating under varying degrees of autonomy. Shared leadership has been defined as a form of distributed leadership where the transference of leadership functions across team members occurs such that member strengths (i.e., knowledge, skills, attitudes, perspectives, contacts) are capitalized on as dictated by either internal or external demands (Burke et al., 2003; Pearce & Sims, 2002--Group Dynamics, Theory, Research, and Practice, 6(2), 172-197.). The current effort is therefore designed to investigate the dynamic relationship between leaders and followers through the theoretical lens of shared or distributed leadership. We aim to investigate the following with respect to long duration, exploration teams: (1) What leadership forms best enable team autonomy; (2) What are the enablers (e.g., team climate, knowledge, behavior, attitudes) to horizontal peer-to-peer leadership?; (3) What is the role that personality occupies in predicting who is likely to take on particular leadership roles and the form of leadership that emerges; and (4) How does this change across the team's life span and autonomy level? Research conducted with both ground based laboratory settings as well as field data collected from analogs is expected to lead to critical information which can support the development of necessary countermeasures for more effectively enabling long-duration distance exploration mission (LDDEM) leadership roles for both ground and flight crews. References Burke, C. S., Fiore, S., & Salas, E. (2003). The role of shared cognition in enabling shared leadership and team adaptability. In J. Conger & C. Pearce (Eds.), Shared leadership: Reframing the hows and whys of leadership (pp. 103-122). Thousand Oaks, CA: Sage Publishers. Pearce, C. L. & Sims, H. P., Jr. (2002). The relative influence of vertical vs. shared leadership on the longitudinal effectiveness of change management teams. Group Dynamics, Theory, Research, and Practice, 6(2), 172-197.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Despite the deleterious effect that stress has on coordinated team interaction (Cannon-Bowers & Salas, 1998), coordination amongst team members is needed to complete mission essential tasks within long duration, distance exploration missions (LDDEM). The isolated nature of LDDEM compounds the effect of stress such that problems may quickly metastasize. This, in turn, may impact team functioning by reducing cohesion and information sharing, situation awareness, and increasing stress as the full capacity of the team is not being utilized for task-based activities. As NASA pushes the edges of space exploration, the degree of autonomy afforded to the portion of the team residing in space is expected to increase due to communication delays and increased distance from Earth (Evans and Ball, 2001). Therefore, it becomes imperative that teams have the leadership capacity to address the temporal dynamics of diversity and other stressors and the resulting rifts that can occur (Jackson et al., 2003). While traditional notions of leadership have focused on leadership residing within a single individual, more recent views of leadership in complex environments argue for leadership being distributed within the team. This does not negate vertical leadership, yet in many environments shared/distributed leadership has been shown to predict team performance over and above vertical leadership (Pearce & Sims, 2002). In our view, vertical leadership sets the conditions for shared leadership to occur. The current effort is designed to investigate the dynamic relationship between leaders and followers through the theoretical lens of shared/distributed leadership. Findings are expected to lead to critical information which can provide insight into (and later support) the development of necessary countermeasures for more effectively enabling LDDEM leadership roles for both ground and flight crews. References Cannon-Bowers, J. A., & Salas, E. (1998). Making decisions under stress: Implications for individual and team training. Washington, DC: APA. Evans Jr., C. H., & Ball, J. R. (Eds.) (2001). Safe passage: Astronaut care for exploration missions. Washington, DC: National Academies Press. Jackson, S., Joshi, A., & Erhardt, N. L. (2003). Recent research on team and organizational diversity: SWOT analysis and implications. Journal of Management, 40(1), 801-830. Pearce, C. L. & Sims, H. P., Jr. (2002). The relative influence of vertical vs. shared leadership on the longitudinal effectiveness of change management teams. Group Dynamics, Theory, Research, and Practice, 6(2), 172-197.
	FINAL REPORTING SUMMARY--JUNE 2018: During this reporting period, three thrusts have been worked on that serve to examine the following research questions: 1) what leadership forms and behaviors best enable team performance in isolated, confined environments; 2) what is the role of time and contextual features in determining the type of team leadership needed; and 3) what are enablers to horizontal peer-to-peer leadership in these type of mission critical teams. The primary progress during this reporting period has related to the analysis of the data collected in earlier project years. Summaries of progress appear below. 1) Longitudinal Data Collection in NASA Analog: The primary focus of analog data collection was to examine team leadership in contexts whereby the team is operating within isolated, confined, and extreme environments (ICE). Analog data collection focused on the nature of team leadership (and what constitutes effective leadership) over time as most of what is known regarding team leadership has occurred outside the context of spaceflight and similar ICE environments and has been static in nature. With regards to team leadership not only were the types of leader behavior investigated, but the degree to which these behaviors were distributed throughout the team (i.e., enacted collectively) and the relation to key team processes and outcomes were examined. Results from the analog indicated that while leadership was a mixture of hierarchical and shared, it was most often distributed through the crew, as seen in both the 14- and 30-day campaigns. Most often, results indicated no single member engaging in leadership functions of a specific type (i.e., action, transition, interpersonal) to the exclusion of

Task Progress:

other team members engaging in the specified group of leadership functions. Results also point to the functionality of sharing leadership. The sharing of transition phase leadership behaviors was significantly correlated with teamwork, cohesion, and effectiveness. Similar results were obtained with respect to the sharing of action phase leadership behaviors. Finally, the sharing of interpersonal leadership behaviors was significantly related to teamwork and cohesion. Trends for the sharing of interpersonal leadership being positively related to team effectiveness were evidenced, but did not reach significance most likely due to low sample size. The functionality of sharing leadership was also examined from a temporal standpoint. To conduct this analysis the data was divided into four quarters based on time. Regressions were then conducted using team effectiveness as the dependent variable and density scores for action, transition, and interpersonal leadership as the independent variable. Results indicated that the distribution of leadership at Time 1 significantly predicted team effectiveness at Time 2. Analyses examining shared leadership at Time 2 predicting team effectiveness at Time 3 indicated results approaching significance. However, other factors seem to be influencing team effectiveness towards the last quarter of the mission as the degree of leadership distribution at Time 3 was not predictive of team effectiveness at Time 4.

2) Laboratory Investigation: The goal of the laboratory investigation was to supplement data collection in the analog environment, where the focus was more on temporal components with respect to team leadership. Efforts with respect to this thrust during the current reporting period, were primarily geared toward data cleaning and analysis, along with the coding of a subset of the team's communication for team leadership behaviors. Results from the laboratory study tended to provide further support for what was being seen in both the analog environment as well as the archival analysis. A few of the emerging findings are highlighted below. First, results suggested that team leadership behaviors were most often distributed throughout the team and not solely enacted by a single individual in the team. Second, incorporating the temporal component we investigated how often different behaviors were performed across missions. Results indicated that over time, the variety of leadership behaviors increased as well as the density of those behaviors (i.e., the degree to which leadership was distributed through the team or shared), with individuals, on average, having a higher density of leadership behaviors in later missions as compared to earlier missions.

We also investigated whether a more objective indicator, participant vocal intensity, might be an appropriate metric for recognizing the degree of shared leadership during particular team performance phases of interaction (i.e., transition, action). Results indicated that the team's vocal intensity was significantly, positively related to shared leadership. However, in the high autonomy condition, this relationship was only true for action oriented shared leadership density and not for transition leadership density. Essentially, these results appear to indicate that talking more during more stressful time periods was likely to be associated with leadership behaviors. However, when autonomy was low (i.e., more structured), vocal intensity was predictive of both transition and action-oriented shared leadership. This has implications in particular in terms of the role of vocal intensity serving as a proxy for measuring leadership during high stress, action heavy performance episodes, but less so for planning phases of similar high stress teamwork contexts.

Finally, analyses were conducted to examine a set of enablers to shared leadership. Results indicated that at the individual level, motivation to lead was significantly predictive of individuals' leadership behavior density, as perceived by their team members. This also held at the team level, where the average of team members' motivation to lead was significantly predictive of shared leadership. However, this was only true for the sub-dimension of affective motivation to lead, implying that shared leadership in this situation may be driven by teams whose members truly enjoy taking on leadership roles. Results also suggested that personality may serve as an enabler to shared leadership.

Results demonstrated that at the team level, average extraversion and openness to experience were significant, positive predictors of shared team leadership, although agreeableness, conscientiousness, and emotional stability were not. This pattern did not significantly change across autonomy or distribution conditions. At the individual level, higher openness to experience was associated with higher individual leadership behavior density, implying that those who were more open to unknown or unfamiliar situations were more likely to step up and take on leadership responsibilities more often. Interestingly, common personality predictors of leadership were not significantly associated with individuals' leadership behavior density ratings from their teammates, including emotional stability, extraversion, and conscientiousness.

3) Analysis of Archival Data: A final thrust of the project was to use a bottoms-up approach to understanding team leadership in isolated, confined environments (ICE). In essence, we employed a mixture of historiometry (Simonton, 1999) and the critical incident technique (Flanagan, 1954) to analyze archived accounts of teams operating in ICE contexts. In doing so, we focused on teams operating in the following ICE contexts – spaceflight, polar exploration, and long duration, round the world, ocean sailboat racing. Using the above methodologies, 152 critical incidents pertaining to team leadership were extracted from 124 data sources consisting of blogs, autobiographies, biographies, and journals. Each critical incident was then independently coded by three subject matter experts with respect to (1) the leadership behavior represented, (2) whether the leadership behavior was enacted by a single individual or multiple individuals (i.e., individual versus shared), (3) formality of leadership (i.e., informal, formal), and (4) locus of leadership (i.e., internal, external). Consensus meetings were held to resolve any discrepancies.

With regard to this reporting period, the primary work on this thrust dealt with diving deeper into the data. In doing so, several key findings emerged. With respect to the team leadership behaviors seen, the predominant number of behaviors identified were in alignment with those argued for within the functional approach to leadership (see Morgeson et al., 2010). While a number of team leadership behaviors emerged, results indicated that the top five team leadership functions included: supporting the social climate, solving problems, structure and planning, sensemaking, and monitoring the team. Furthermore, structure and planning and sensemaking were found to collectively account for 74% of the leadership functions seen during the teams' transition phases, while team problem solving, supporting the social climate, and monitoring the team collectively accounted for 70% of the action phase functions seen.

Additionally, our data allowed examination of the structural aspects of leadership in extreme teams. Taking a functional perspective to leadership we moved past solely focusing on a leader-centric view of leadership to one which represents a combination of team and leader-centric approaches. Results found evidence for leadership functions being enacted through a combination of leadership distributed throughout the team as well as leadership being enacted by a single individual. Furthermore, results suggested that leadership in extreme teams was enacted by both formal and informal leaders as well as formal and informal leaders working together to enact leadership. This later result is reminiscent of work on high reliability organizations, but brought down to the team level. Both of the above results suggest that in comprising teams to work within such extreme environments attention needs to be paid to selecting individuals who can alternate between leadership and followership. In addition, formal leaders should be taught to promote a climate where the source of leadership is not limited to high status individuals, but allowed to flow from all members on the team. In this way serving to build the leadership capacity within the team. Finally, findings also begin to point to the importance

of examining the specific functions that comprise transition and action phases. For example, results suggested that certain leadership functions were most often enacted through a mixture of formal, informal, and joint mechanisms; whereas other functions were most typically enacted through formal mechanisms. Differences were also seen in the degree to which specific functions tended to be enacted in a leader or team-centric approach.

References

Simonton, D. (1998). A history of European women's work: 1700 to the present. Taylor & Francis US. p. 269.

Flanagan, J. (1954). The critical incident technique. Psychological Bulletin, 51, 327-358.

Morgeson, F. P., DeRue, D. S., & Karam, E. P. (2010). Leadership in Teams: A Functional Approach to Understanding Leadership Structures and Processes. Journal of Management, 36, 5-39.

ANNUAL REPORTING FOR AUGUST 2016-JULY 2017, AS OF JULY 2017:

The work within the grant can be divided into three thrusts: laboratory experimentation, analysis of archival analysis, and data collection in analog environments. Our progress within the current reporting period for each of the three is detailed below:

Laboratory Experimentation. Our data collection efforts for the laboratory experimentation are now complete. We collected data from approximately 278 4-person teams who completed three missions within the context of a space-simulation. Independent variables that were manipulated include degree of autonomy and team member distribution. Of interest is the degree to which leadership behaviors were distributed throughout the team and how team autonomy (and changes thereof) impacted the distribution and importance of particular behaviors. The end goal is to determine the team leadership behaviors and particular forms of these behaviors which are the most functional under differing levels of autonomy. In this vein, the relationship between leadership density and key team process and outcome variables are being examined (e.g., cohesion, teamwork quality, team effectiveness). Within the current reporting period, we have worked to prepare the self-report and objective measures included in the study for analysis and run preliminary analyses on a subset of key variables. Additionally, a majority of our efforts have been on training a set of six raters to code the team's communication with respect to the enactment of team leadership behaviors.

Archival Analysis. A second thrust of this project was to use a modified historiometric approach to better understand what functional leadership may look like in isolated, confined environments (i.e., spaceflight and similar environments). Last year we completed our analysis and began to write up our results. Within the current reporting period, we have presented these results at conferences and have submitted a manuscript to a journal based on the findings. Results indicate a mixture of shared and hierarchical leadership and highlight the ways in which leadership is most commonly manifested.

Human Exploration Research Analog (HERA). The final thrust of this project related to data collection in an analog environment. In this vein, we continued collecting data in HERA. HERA is a NASA mission simulation located out at Johnson Space Center. HERA simulates a 2-year asteroid exploration mission. It involves a 4-member crew, selected from NASA volunteers. The missions are conducted 4 times a year and crews are isolated and confined to the HERA analog during the entirety of the mission. We have collected data in Campaign 2 (14-day mission) and Campaign 3 (30-day mission). During this reporting period, data collection for Campaign 3 concluded. Our primary efforts with this reporting period have been on analyzing the primary data that was collected during Campaigns 2 and 3 (crew reported measures), coordinating data sharing agreements with those whom we are either sending or receiving data from, and identifying a subset of the crew's communication to code for instances of leadership behavior. As part of this later effort, we have developed a codebook and trained three raters. Raters have coded approximately 62 files which cut across the temporal span of the mission. These ratings not only allow a more in-depth examination of the leadership behaviors than what can be gained through the self-report measures, but will later be examined in relation to key antecedents and key team outcomes. Initial findings have indicated trends for leadership being distributed throughout the crew, its relation to key team outcomes, and suggest that shared leadership is differentially related to key outcomes across time (i.e., time matters). Initial results from the communication coding have indicated that specific leadership behaviors vary in the degree to which they are distributed throughout the team and are impacted by stressors. Communication coding is still in progress in an effort to sample across a subset of the operational tasks the crews perform.

Bibliography Type:	Description: (Last Updated: 07/15/2026)
Articles in Peer-reviewed Journals	Burke CS, Shuffler ML, Wiese CW. "Examining the behavioral and structural characteristics of team leadership in extreme environments." J Organ Behav. 2018 Jul;39(6):716-30. https://doi.org/10.1002/job.2290 , Jul-2018
Books/Book Chapters	Burke CS, Georganta E, Hernandez C. "The importance of time in team leadership research." in "Team dynamics over time." Ed. E. Salas, W.B. Vessey, L.B. Landon. Bingley, UK: Emerald Publishing, 2017. p. 95-122. https://doi.org/10.1108/S1534-085620160000018005 , Aug-2017
Books/Book Chapters	Flynn ME, Verhoven DC, Shuffler ML. "Temporal dynamics in multiteam systems: An integrative perspective for future research and practice." in "Team dynamics over time." Ed. E. Salas, W.V. Vessey, L.B. Landon. Bingley, UK: Emerald Publishing, 2017. p. 287-322. https://doi.org/10.1108/S1534-085620160000018012 , Aug-2017
Books/Book Chapters	Verhoeven DC, Cooper T, Flynn M, Shuffler ML. "Transnational team effectiveness." in "The Wiley-Blackwell Handbook of the Psychology of Team Working and Collaborative Processes." Ed. E. Salas, R. Rico, J. Passmore. Malden, MA: Wiley-Blackwell, 2017. p. 73-102. https://doi.org/10.1002/9781118909997.ch4 , Sep-2017